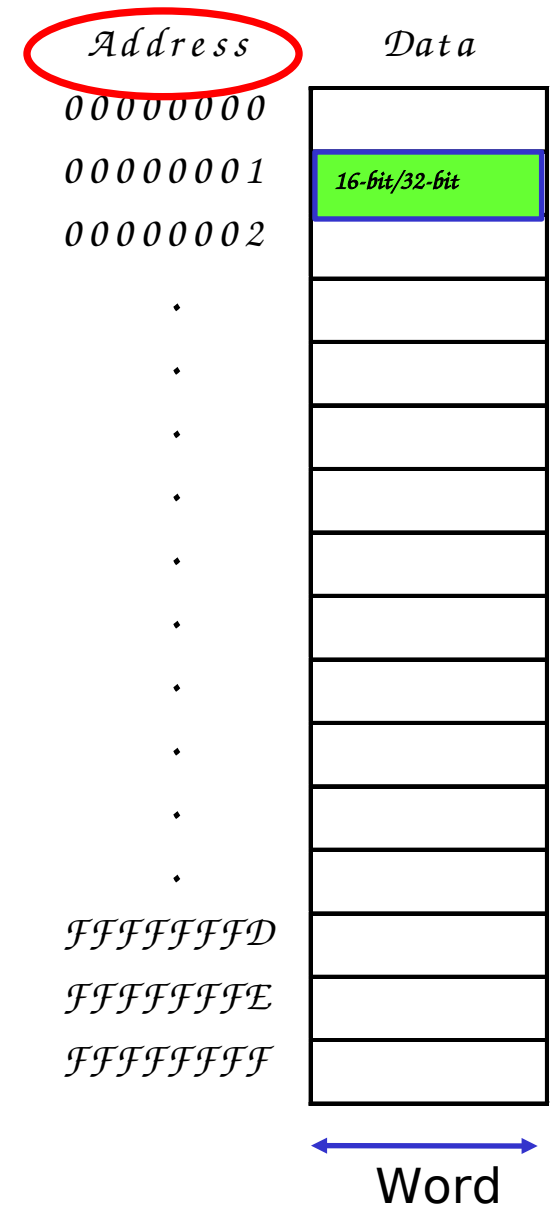


Memory: Introduction

Picture of Memory

- You can think of memory as being one big array of data.
 - The **address** serves as an array index.
 - Each address refers to one word of data.
- You can read or modify the data at any given memory address, just like you can read or modify the contents of an array at any given index.



Memory Signal Types

- Memory signals fall into three groups
 - Address bus - selects one of memory locations
 - Data bus
 - Read: the selected location's stored data is put on the data bus
 - Write (RAM): The data on the data bus is stored into selected locations
 - Control Signals - specifies what the memory is to do
 - Control signals are usually active low
 - Most common signals are:
 - CS: Chip Select; must be active to do anything
 - OE: Output Enable; active to read data
 - WR: Write; active to write data

Random Access Memory

Random Access Memories

- So called because it **takes the same amount of time** to address any particular part
- Types of RAM
 1. Static RAM (SRAM), fast, expensive
 2. Dynamic RAM (DRAM), slow, cheap

1K x 16 memory

- Variety of sizes
 - From 1-bit wide

Memory address		Memory contents
Binary	Decimal	
0000000000	0	10110101 01011100
0000000001	1	10101011 10001001
0000000010	2	00001101 01000110
	⋮	⋮
	⋮	⋮
	⋮	⋮
	⋮	⋮
1111111101	1021	10011101 00010101
1111111110	1022	00001101 00011110
1111111111	1023	11011110 00100100

- Memory size specified in bytes
 - 1K x 16 bit □ 2KB memory
- 10 address lines and 16 data lines

Memory: Writing

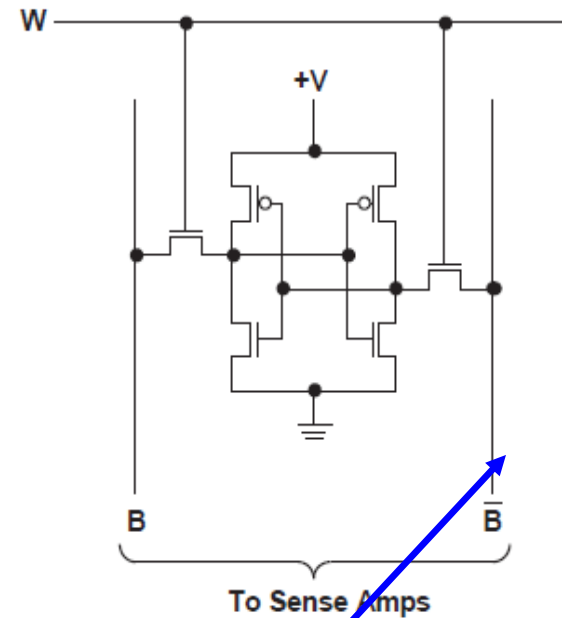
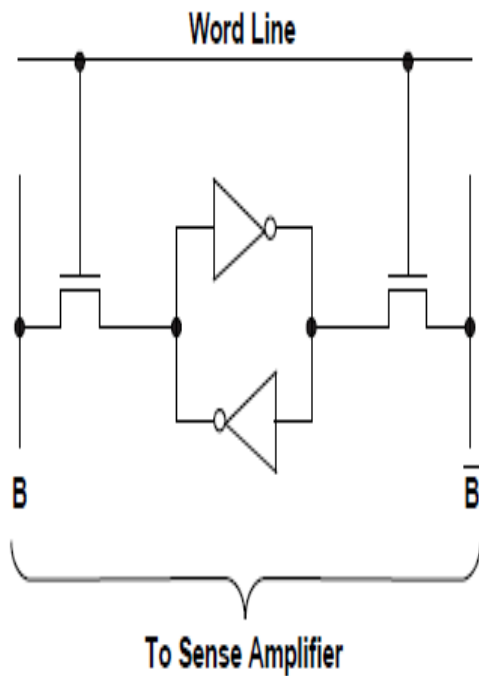
- Sequence of steps
 - ▣ Setup address lines
 - ▣ Setup data lines
 - ▣ Activate write line (maybe a pos edge)
- The write cycle time is the maximum time from the application of the address to the completion of all internal memory operations required to store a word.

Memory Reading

- Steps
 - Setup address lines
 - Activate read line
 - Data available *after specified amount of time*
- Read cycle usually is shorter than write cycle.

Static RAM

Static RAM: 4T and 6T

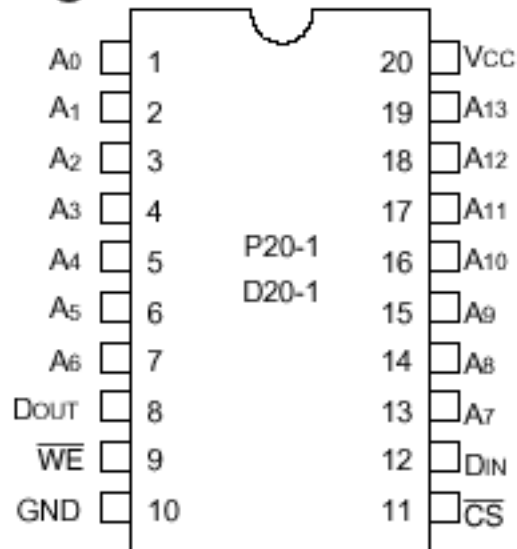


Six Transistors to implement a memory cell

Pin Diagram

IDT6167SA/LA
CMOS Static RAM 16K (16K x 1-Bit)

Pin Configurations



DIP Top View

2981 dnr 02

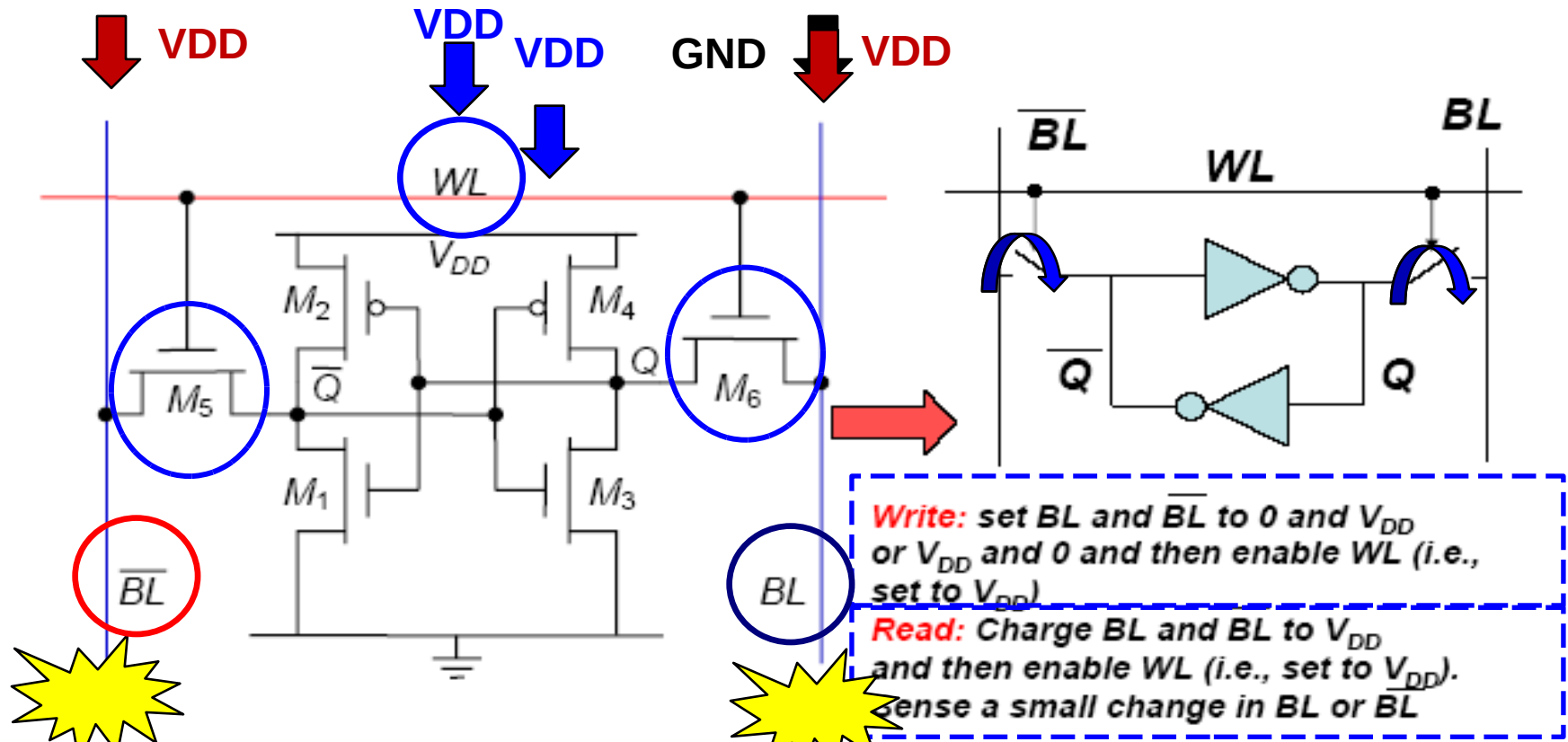
Pin Descriptions

Name	Description
A ₀ - A ₁₃	Address Inputs
$\overline{CS}$	Chip Select
$\overline{WE}$	Write Enable
V _{CC}	Power
D _{IN}	DATA _{IN}
D _{OUT}	DATA _{OUT}
GND	Ground

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SRAM Cell



- State held by cross-coupled inverters (M_1 - M_4)
- Retains state as long as power supply turned on
- Feedback must be overdriven to write into the memory

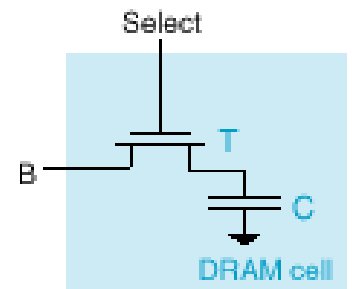
Dynamic RAM

Dynamic Memory

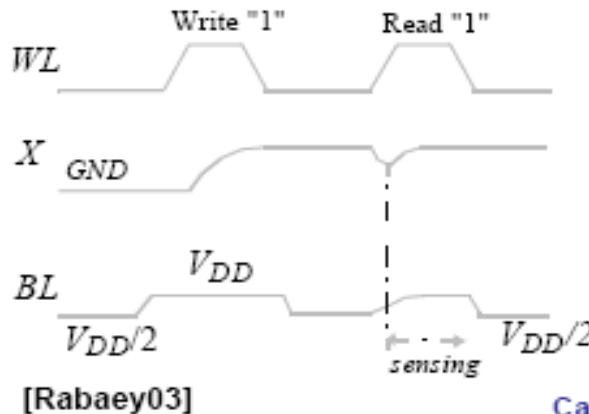
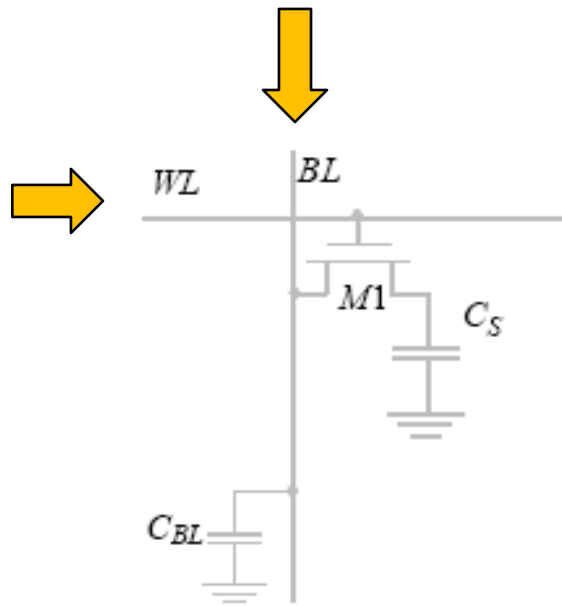
- **Dynamic memory** is built with capacitors.
 - A stored charge on the capacitor represents a logical 1.
 - No charge represents a logic 0.
- However, capacitors lose their charge after a few milliseconds. The memory requires constant **refreshing** to recharge the capacitors. (That's what's "dynamic" about it.)
- Dynamic RAMs tend to be physically smaller than static RAMs.
 - A single bit of data can be stored with just **one capacitor and one transistor**, while **static RAM cells typically require 4-6 transistors**.
 - This means dynamic RAM is **cheaper** and **denser**—more bits can be stored in the same physical area.

Dynamic RAM

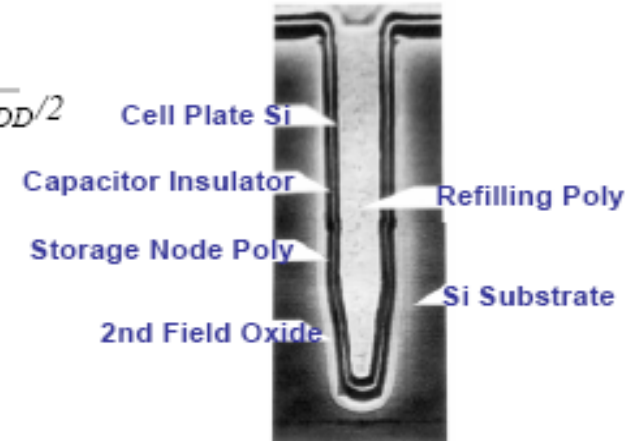
- Capacitor can hold charge
- Transistor acts as gate
 - ❑ No charge is a 0
 - ❑ Can close switch & add charge to store a 1
- Then open switch (disconnect)



DRAM Cell



**DRAM uses
Special
Capacitor
Structures**



To Write: set Bit Line (BL) to 0 or V_{DD}
& enable Word Line (WL) (i.e., set to V_{DD})

To Read: set Bit Line (BL) to $V_{DD}/2$
& enable Word Line (i.e., set it to V_{DD})

- **DRAM relies on charge stored in a capacitor to hold state**
- **Found in all high density memories (one bit/transistor)**
- **Must be “refreshed” or state will be lost – high overhead**

DRAM Characteristics (Why Slow!)

o Destructive Read

- ❑ When cell read, charge removed
- ❑ Charge must be restored after a read

o Refresh

- ❑ Capacitors are not perfect! there's steady leakage
- ❑ Charge must be restored periodically

o DRAM **are dense** (lots of cells) so there are **many address lines**.

- ❑ To reduce the physical size of DRAM we can reduce the number of pins by applying the address lines serially in two parts:
 - Row Address, and then
 - Column Address

How DRAM Works

